AP CALCULUS AB · UNIT 1

1.16 Working with the Intermediate Value Theorem (IVT)

Lesson Overview

- State every hypothesis of the Intermediate Value Theorem.
- Use endpoint values to guarantee roots or other target outputs.
- Count minimum distinct solutions from disjoint sign-changing intervals.
- Use auxiliary functions and bisection to localize guaranteed solutions.

Key Knowledge and Vocabulary

Closed interval. The function must be continuous on $[a, b]$.

Target bracket. The desired value N must lie between $f(a)$ and $f(b)$.

Conclusion. There exists at least one $c \in (a, b)$ such that $f(c) = N$.

Existence, not uniqueness. IVT guarantees at least one solution, not exactly one.

Auxiliary function. Rewrite $F(x) = G(x)$ as $H(x) = F(x) - G(x) = 0$ and apply IVT to H.

Explanation and Strategy

Step 1. State continuity, endpoint values, the bracketing inequality, and the existence conclusion.
Step 2. Use disjoint intervals to guarantee distinct solutions.
Step 3. For bisection, retain the half-interval whose endpoint values still bracket the target.

Solved Examples

Worked Example: Guarantee a root

Show $x^3 + x - 1 = 0$ has a solution in $(0, 1)$.

Solution. Let $f(x) = x^3 + x - 1$, continuous on $[0, 1]$. Since $f(0) = -1$ and $f(1) = 1$, 0 lies between the endpoint values. IVT guarantees $c \in (0, 1)$ with $f(c) = 0$.

Worked Example: Nonzero target

If f is continuous on $[2, 5]$, $f(2) = 1$, and $f(5) = 9$, show $f(c) = 6$ for some c.

Solution. 6 lies between 1 and 9, so IVT guarantees at least one $c \in (2, 5)$ with $f(c) = 6$.

Worked Example: Intersection

Show $e^{-x} = x$ has a solution in $(0, 1)$.

Solution. Let $H(x) = e^{-x} - x$, continuous on $[0, 1]$. $H(0) = 1 > 0$ and $H(1) = e^{-1} - 1 < 0$, so IVT guarantees a zero and hence an intersection.

Leveled Practice

Shared Representation / Data

x	0	1	2	3	4
$f(x)$	3	-1	2	5	-2

Easy Level

Foundational fluency. Focus on notation, definitions, and direct procedures.

1. List the three main IVT hypotheses/conclusion components.

2. If f is continuous on $[1, 4]$, $f(1) = -2$, and $f(4) = 5$, what value is guaranteed to occur: -3, 0, or 8?

3. Does IVT guarantee uniqueness?

4. Why is continuity essential?

Medium Level

Apply the idea in one or two connected steps.

5. Show $x^2 - 3 = 0$ has a solution in $(1, 2)$.

6. Use the shared table to identify intervals that guarantee a root.

7. If f is continuous and $f(a) = 7$, $f(b) = 2$, is a point with $f(c) = 5$ guaranteed?

8. If $f(a) = f(b) = 1$, can IVT guarantee a zero between them?

Hard Level

Connect representations, conditions, and justification.

9. Show $\cos x = x$ has a solution in $(0, 1)$.

10. Use the shared table to give the minimum number of distinct roots guaranteed on $(0, 4)$.

11. A student says the table guarantees exactly three roots. Correct the claim.

12. Show $x^5 + 2x + 4 = 0$ has a root in $(-2, -1)$.

Difficult Level

Use nonroutine structure and precise mathematical reasoning.

13. Let f be continuous on $[0, 6]$ with $f(0) = 2$, $f(2) = -1$, $f(4) = 3$, $f(6) = -4$. What minimum number of roots is guaranteed?

14. Show that every continuous function $f : [0, 1] \to [0, 1]$ has a fixed point.

15. If f is continuous on $[a, b]$ and never equals 0, prove that it cannot take both positive and negative values.

Challenge Level

Synthesize, construct, generalize, or prove.

16. Prove that any odd-degree real polynomial has at least one real root.

17. A continuous function satisfies $f(0) = 0$, $f(1) = 2$, $f(2) = 0$. Does IVT guarantee a point where $f(c) = 1$ in both $(0, 1)$ and $(1, 2)$?

AP-Style Practice

Multiple-Choice Practice – No Calculator

Choose the best answer. Use exact values unless a decimal approximation is requested.

1. Which condition is required by IVT?

(A) Differentiability on (a, b)

(B) Continuity on $[a, b]$

(C) $f(a) = f(b)$

(D) A unique root

2. If f is continuous, $f(2) = -3$, and $f(5) = 4$, IVT guarantees

(A) exactly one root

(B) at least one root in $(2, 5)$

(C) a root at an endpoint only

(D) no root

3. Which conclusion is too strong?

(A) At least one solution exists.

(B) A target value is attained.

(C) Exactly one solution exists.

(D) A solution lies in the open interval.

Multiple-Choice Practice – Calculator Required

Choose the best answer. Use exact values unless a decimal approximation is requested.

4. A calculator gives $f(1.2) = -0.4$ and $f(1.3) = 0.2$ for a continuous f. What is guaranteed?

(A) A zero in $(1.2, 1.3)$

(B) Exactly one zero

(C) A maximum

(D) A discontinuity

5. Bisection begins with a continuous function satisfying $f(2) < 0 < f(3)$ and finds $f(2.5) > 0$. Which interval still guarantees a root?

(A) $(2, 2.5)$

(B) $(2.5, 3)$

(C) $(2, 3)$ only, not a half

(D) No interval

Free-Response Practice – No Calculator

Let $f(x) = x^3 - 2x - 2$.

(a) Use IVT to prove that f has a zero in $(1, 2)$. [3 points]

(b) Use one bisection step to choose either $(1, 1.5)$ or $(1.5, 2)$ as a smaller guaranteed interval. [2 points]

(c) Explain why IVT alone does not prove the root is unique. [2 points]

Free-Response Practice – Calculator Required

Use a calculator to localize a solution of $e^{-x} = x$.

(a) Define an auxiliary function and find a sign-changing interval of length 0.1. [2 points]

(b) Use bisection or a solver to approximate the solution to three decimals. [2 points]

(c) State precisely what continuity theorem justifies the existence before the numerical approximation. [2 points]

END OF UNIT 1 WORKBOOK

Review unresolved errors, rewrite weak justifications, and retake selected AP-style items without notes.